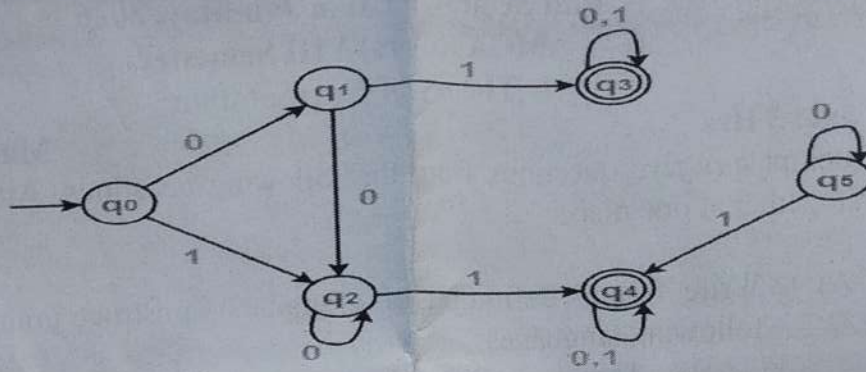
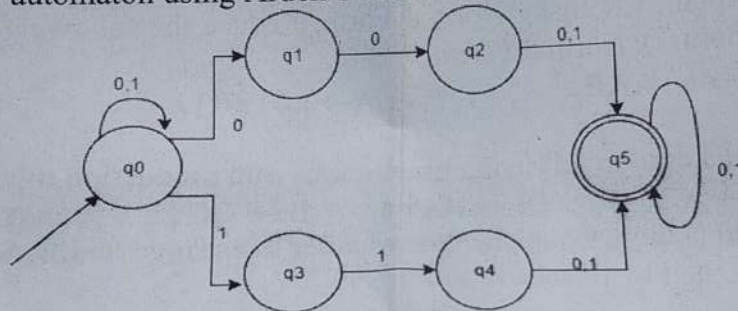


- B) Construct a minimum state automaton equivalent to the finite automaton described by the following transition graph: (6)



5. A) Construct finite automata equivalent to following regular expressions: (3+3)
- i) $(ab + a)^* (aa + b)$ ii) $a(a + b)^* ab$

- B) Construct a regular expression equivalent to following finite automaton using Arden's Theorem: (6)



6. A) Write formal definition of Pushdown Automata (PDA). Construct a Pushdown Automata (PDA) for following language: (3+3)
- $L = \{ \{0^n 1^n \mid n \geq 1\} \}$

- B) Construct a Pushdown Automata (PDA) for following grammar: (6)
- $S \rightarrow aAB \mid bBA \quad A \rightarrow bS \mid a \mid aS \quad B \rightarrow b$
- Check whether the string **abaabb** is accepted by the PDA. Show stack's contents at each instance.

7. A) Give formal definition of Turing machine. Construct a Turing Machine for following language: (2+4)
- $L = \{0^n 1^n 2^n \mid n \geq 1\}$

- B) Define the following: (2x3)
- i) Church Turing Thesis
- ii) Recursively Enumerable Language
- iii) Recursive Language

IC-2K22-20

IC-811B

International Institute of Professional Studies, DAVV
End-Semester Exam April - 2026
MCA (5 Years) VIII Semester, Section A & B
Data Mining and Warehousing

Time:-3 Hrs

Max.Marks:60

Note: -Attempt any five questions; Solve all parts of question at the same place.

- Q1. A What are the essential differences between the MOLAP and ROLAP models? Also list a few similarities. 6
- B Describe the different classifications of Association rule mining. 6
- Q2. C Explain the operation of data-cube with suitable examples. 6
- D Explain the differences between Knowledge discovery and data mining. 6
- Q3. E Entropy is an important concept in information theory. Explain its significance in mining context. 6
- F Explain the algorithm for constructing a decision tree from training samples. 6
- Q4. G Explain the cluster analysis methods briefly. 6
- H How is data warehouse different from a database? How are they similar? 6
- Q5. I Can you briefly describe the four stages of knowledge discovery (KDD). 4
- J Consider the dataset ID. Given the minimum support=2, apply the apriori algorithm on this dataset. 8

Transaction ID	Items
100	A,C,E
200	B,C,E
300	A,B,C,E
400	B,E,A

- Q6. K How density based methods are used for clustering? Explain. 4

- L Dataset (Play football Example),
Solve using the Hunt algorithms, and create decision tree

Outlook	Temperature	Humidity	Wind	Play
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No

- Q7. M Briefly describe Web Mining Process.

4

- N Difference between Web Content, Structure and Usage Mining.

8

Roll No. IC-2K22-20

IC-802B/IT805B

International Institute of Professional Studies, DAVV
End-Semester Exam Jan-May, 2026
MCA (5 Years)/ M. Tech (5 Year) VIII Sem
Enterprises Computing Technique

Duration : 3 hrs

Max. Marks 60

Note: Answer any five questions. Make suitable assumption if any

- 1 a. List the things to consider while building large business application. 6
- b. With diagram explain Implicit and explicit middleware. 6
- 2 a. With block diagram explain the working of EJB. 6
- b. Write a session bean to calculate the sum of given three number. Note 6
write all required interface, class and deployment descriptor.
- 3 a. Write the client code for question 2b. 6
- b. Explain the following term 6
 - I. What happens during Create and remove of session bean
 - II. Parameter passing in RMI
 - III. Concept of instance pooling in State full session bean.
- 4 a. Write issue in entity bean instance are view into a databases. 6
- b. Write notes on finder method of entity bean. 6
- 5 a. Write note on the use of getPrimaryKey() method . 6
- b. With diagram explain the CMP entity bean. 6
- 6 Write an entity bean for Student bean having following detail. Note: write 12
the required class, interfaces and deployment descriptor.

<<Student>>	
rollNumber	
name	
Class	
dept	
findByName():student	
findByDept():Employeeelist	
findTotalNoOfstudent in Dept(Dept)	
Note: other required method	

7

Write notes on any three

12

- I. JMS API.
- II. Transactional Attribute.
- III. Poison message in message driven bean.
- IV. Explain how 1:M relationship is achieved in CMP

107-26

IC-812A

Roll No. IC-2K22-20

International Institute of Professional Studies
MCA(5Yrs) -VIII Semester
End-Sem Examination April-May, 2026
IC-812A: Soft Computing

Time: 3 Hrs

Max. Marks: 60

NOTE: All questions are compulsory. Attempt any two parts from each question. All questions carry equal marks. Answers must be brief and to the point. Attempt all the parts of a question on same place. Any assumptions made answering the questions should clearly be stated.

- Q.1 (a) Give a definition of soft computing and discuss how it can be used to solve practical issues. Distinguish between Hard Computing and Soft Computing according to their traits and methods of solving problems. 6
- (b) Differentiate between the "Binary or Crisp Logic" used in hard computing and the logic systems used in soft computing. 6
- (c) Explain the structure of a Rule-Based Expert System and the role of its components. 6
- Q.2 (a) What are the implications of using hedges in fuzzy logic, and how do they enhance the representation of linguistic variables? 6
- (b) Define and explain with a suitable diagram the significance of the Core, Support, and Crossover Point of a fuzzy membership function 6
- (c) Describe and draw the **Linear Representation Rose** and **Linear Representation Down** membership functions. Consider the fuzzy membership function for "young":
 $\text{young}(x) = 1$, if $\text{age} \leq 20$
 $\text{young}(x) = (30 - \text{age})/10$, if $20 < \text{age} \leq 30$
 $\text{young}(x) = 0$, if $\text{age} > 30$
 Calculate the membership value for ages:
 (a) 18, (b) 25, (c) 32, (d) 27. 6
- Q.3 (a) Explain with neat block diagram the various components and operation of a fuzzy inference system. 6
- (b) With the help of necessary block diagrams, compare Mamdani and Sugeno Fuzzy Inference Systems. 6
- (c) Explain with a suitable diagram and write properties of Trapezoidal membership function. A fuzzy set "Middle Aged" is defined by $\text{TrapMF}(x; 30, 40, 55, 65)$. Calculate the membership grade for an individual who is 35 years old. 6
- Q.4 (a) Describe the structural components of a biological neuron (dendrites, soma, axon) and explain how they are functionally mapped to an Artificial Neural Network (ANN). 6
- (b) What is mean by learning what is supervised and unsupervised learning give example of each. 6
- (c) Describe the two distinct passes (forward and backward) involved in the Backpropagation algorithm. 6
- Q.5 (a) Define Genetic Algorithms (GA) and identify the natural process they are designed to simulate. 6
- (b) Briefly explain the significance of the Schema Theorem in understanding how GAs work. 6
- (c) Explain how GAs handles the problem of getting stuck in local optima compared to traditional calculus-based methods. 6

IC-2K22-20

IC-801B

International Institute of Professional Studies
End Semester Exam April-May 2026
MCA(5yr) VIII Semester
Mobile and Wireless Computing

Time: 3 hrs

Max. Marks :60

Note: Attempt any five questions.

- | | | |
|------|--|---|
| Q1a) | Explain features of different generation of mobile communication. | 6 |
| b) | What is the spread spectrum? Explain DSSS and FHSS. | 6 |
| Q2a) | Explain MACA demonstrating how it solved hidden and expose terminal problem. | 6 |
| b) | Explain CDMA, its advantaged and disadvantages, | 6 |
| Q3a) | Define- frequency reuse, cell splitting, capacity of cell, and cell sectoring in cellular system with diagram. | 6 |
| b) | Explain different hand-off strategies and types of handovers. | 6 |
| Q4a) | Explain GSM architecture in detail. | 6 |
| b) | Describe the authentication and security mechanisms in GSM. | 6 |
| Q5a) | Explain IEEE 802.11 layers and functions. | 6 |
| b) | Explain Bluetooth. Explain different states of a Bluetooth device. | 6 |
| Q6a) | Discuss the Mobile Node, Home Agent, Foreign Agent, Care of address in Mobile IP in detail. | 6 |
| b) | What are the major challenges in routing in MANETs? Explain issues like dynamic topology, limited bandwidth, and energy constraints. | 6 |
| Q7a) | Explain and compare indirect TCP and snooping TCP. | 6 |
| b) | Discuss the architecture of the Internet of Things (IoT). Explain its layers and enabling technologies. | 6 |

IC-2K22-20

IC-812

International Institute of Professional Studies
End Semester Exam Jan-May, 2026
MCA (5 Yrs) VIII Semester
Theory of Computation

Time: 3 Hrs**Max. Marks: 60**

Note: Attempt *any five* questions from the following questions. Attempt subparts of a question together at one place.

1. A) Write formal definition of grammar. Construct grammar for the following languages: (2+4)
 - i) $L_1 = \{\text{set of all palindrome strings over } \{a,b\}^*\}$
 - ii) $L_2 = \{\text{set of legally balanced parenthesis over } \{(,)\}^*\}$
- B) What do you mean by ambiguous grammar? Prove that the grammar with following production rules is ambiguous: - (2+4)

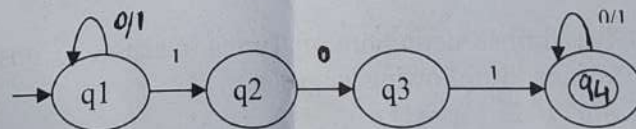
$$S \rightarrow a \mid abSb \mid aAb \quad A \rightarrow bS \mid aAAb$$
2. A) Explain Chomsky Normal Form. Reduce the following grammar to Chomsky Normal Form: (2+4)

$$S \rightarrow a \mid AAB \quad A \rightarrow ab \mid aB \mid \lambda \quad B \rightarrow aba \mid \lambda$$
- B) Consider the following grammar G with production rules: (3+3)

$$S \rightarrow AB \mid BC \quad A \rightarrow BA \mid a \quad B \rightarrow CC \mid b \quad C \rightarrow AB \mid a$$
 Find using CKY algorithm whether the strings **baaba** and **baaab** are accepted by the above grammar
3. A) Construct Deterministic Finite Automata for following languages: (3+3)

$$L_1 = \{\text{set of all strings over } \{a,b\}^* \text{ that ends with } aba\}$$

$$L_2 = \{\text{set of all strings over } \{a,b\}^* \text{ that ends in } b \text{ and does not contain substring } aa\}$$
- B) Write the conditions for automata to be non-deterministic. Convert the following non-deterministic finite automata (NFA) into deterministic finite automata (DFA):- (3+3)



4. A) Write the formal definition of Mealy machine and Moore machine. Construct a Moore machine equivalent to the following Mealy machine:- (3+3)

Present State	a=0		a=1	
	State	Output	State	Output
→q1	q1	1	q2	0
q2	q4	1	q4	1
q3	q2	1	q3	1
q4	q3	0	q1	1

P.T.O.